

Assignment: Comparative Study of Pretrained CNN Models for Binary Image Classification

Objective

The goal of this assignment is to help you understand, implement, and critically analyze different pretrained convolutional neural network (CNN) models. You will also develop essential research skills including experimentation, technical writing, and comparative analysis.

Task Overview

1. Select a dataset suitable for binary image classification.
2. Choose 5 pretrained models including MobileNet, ResNet and others
3. Train and evaluate all models using consistent preprocessing.
4. Use transfer learning (fine-tuning optional but encouraged).

Experiments and Analysis

For each model, report performance on training, validation, and test sets. Include metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix. Also include training curves (loss and accuracy).

Report Requirements

1. Introduction: overview of image classification and transfer learning.
2. Background: architecture, parameters, and CNN blocks for each model.
3. Dataset Description: source, size, class distribution.
4. Experimental Setup: preprocessing, splits, hyperparameters.
5. Results: tables and figures for all models.
6. Comparative Analysis: performance, complexity, and observations.
7. Discussion: insights, challenges, limitations.
8. Conclusion: summary and future work.

Submission Requirements

1. PDF report.
2. Jupyter Notebook / code (well organized and commented).

Grading Criteria

Dataset selection and analysis: 10

Model implementation: 20

Experimental setup: 15

Results & visualization: 15

Comparative analysis: 15

Discussion: 10

Report quality: 10

Code quality: 5

Total: 100

Notes

Focus on understanding, not just execution. Write clearly and professionally. Avoid plagiarism.